

ASSIGNMENTS

Machine Learning Algorithms with C++ (CSE 3374)



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Major Assignment

1. A company wants to predict advertising sales based on:

x1: TV advertisement budget

x2: Online advertisement budget

They collect the following data:

TV(x1)	Online(x2)	Sales(y)
1000	2000	8000
2000	1000	7000
3000	3000	14000

The company decides to use multiple linear regression to model the relationship between the independent variables and the dependent variable.

(a) Construct the design matrix X and target vector y .

(b) Find the parameter vector β .

2. Apply the K-Means clustering algorithm with $K = 3$ to the following dataset:

Data Points	x1	x2
P1	2	10
P2	2	5
P3	8	4
P4	5	8
P5	7	5
P6	6	4

3. The data set of five two dimensional data points is given below.

Data Points	x1	x2
P1	1	1
P2	2	1
P3	4	3
P4	5	4
P5	3	2

(a) Compute the Euclidean distance matrix for all pairs of points.

(b) Apply Agglomerative Hierarchical Clustering using Single Linkage method.

(c) Draw the dendrogram (hierarchical tree) representing the clustering process.